

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil Engineering) SEMESTER - 6 Summer 2025 (Regular)

Course :Bachelor of Technology (Civil Engineering) Branch : {FACULTYNAME}

Semester : SEMESTER - 6

Subject Code & Name: BTCVC601 - DESIGN OF RC STRUCTURES

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
-

Q1. Objective type questions. (Compulsory Question)

12

While designing a beam, what points are considered from the following ?

- a. Bending Moment
- b. Shear Force
- c. Neither Shear force nor Bending moment
- d. Both Shear force and Bending moment

What is the ratio of the strength of material to the permissible stress is called?

- a. Factor of Safety
- b. Shear factor
- c. Creep factor
- d. Strength Modulus

3 The concrete in the _____ zone should not be taken into account while determining the neutral axis.

- a. Compression
- b. Neutral Zone
- c. Centroidal Zone
- d. Tension Zone

4 Which of the following is the function of the transverse reinforcement in a column?

- a. To prevent longitudinal buckling of longitudinal reinforcement.
- b. To prevent certain brittle failure.
- c. To impart certain ductility to the column.
- d. To reduce effect of creep

5 A column is 5m long (effective) and 500mm in diameter. What kind of column is this?

- a. long
- b. Slender
- c. Short
- d. Thick

- 6 What is the horizontal distance or spacing between the reinforcement bars of an 80mm thick slab?
- 340mm
 - 440mm
 - 240mm
 - 540mm

- 7 What is the not function of the transverse reinforcement?
- It distributes the load more evenly and uniformly on the slab
 - It prevents the shrinkage or temperature effect
 - It keeps the main reinforcement in position
 - It works as the main reinforcement

- 8 What is the objective of providing foundation to a structure?
- To provide a base to the structure
 - To stabilize the soil below the structure
 - To distribute the load to the soil
 - For the compaction of the soil below the structure

- 9 Which of the following is not an assumption for working stress design method?
- Tensile strength of concrete is considered
 - Bond between steel and concrete is perfect within the elastic limit of steel
 - Concrete is elastic
 - A section which is plane before bending remains plane after bending

- 10 The stress strain curve of concrete as per IS-456 is
- Perfect straight line up to failure
 - Straight line up to 0.002 strain value and then parabolic up to failure
 - Parabolic up to 0.002 strain value and then uniform up to failure
 - Linear up to 0.002 strain value and then uniform up to failure

- 11 Which of the following is not a limit state of serviceability?
- Deflection
 - Cracking
 - Torsion
 - Durability

- 12 Limiting moment of Resistance of R.C. beam for Fe415 grade steel is
- $M_{u\lim} = 0.138 F_{ck} b d^2$
 - $M_{u\lim} = 0.148 F_{ck} b d^2$
 - $M_{u\lim} = 0.133 F_{ck} b d^2$
 - $M_{u\lim} = 0.130 F_{ck} b d^2$

Q2. Solve the following.

- A) Differentiate between Under-Reinforced and Over –Reinforced Section 6
- B) Find the moment of resistance of RC beam 250mm wide and 500 mm effective 6 depth. The area of tensile steel is 1200 mm^2 . Also calculate the actual stresses in concrete and steel at the maximum moment of resistance, if permissible stresses in steel and concrete are 140 N/mm^2 and 5 N/mm^2 respectively. $m = 18.66$

Q3. Solve the following.

- A) Write about advantages of LSM over WSM 6
- B) Derive the stress block parameters for a singly reinforced concrete beam under flexure 6

Q4. Solve Any Two of the following.

- A) Singly reinforced beam $250 \text{ mm} \times 500 \text{ mm}$ in section is reinforced with 4 bars of 16 mm diameter with an effective cover of 25 mm. Effective span of the beam is 5 m. Assuming M20 concrete and Fe 415 steel, determine the central concentrated load P that can be carried by the beam in addition to its self-weight. Use LSM.
- B) Design a simple supported singly reinforced beam of 230 mm wide which carries a factored load of 30 kN/m including its self-weight. The span of beam is 6m. Design the beam for flexure and shear by limit state method if materials are M25 concrete and HYSD bars of Fe415.
- C) A T-beam of effective flange width of 1800 mm, thickness of slab 100 mm, width of rib 230 mm and effective depth of 500 mm is reinforced with 4 nos. 25 mm diameter bars. Calculate the factored moment of resistance if M20 and Fe500 is used. Use LSM.

Q5. Solve Any Two of the following.

- A) A rectangular beam is restricted to a depth of 350 mm due to architectural constraints. Design the beam section to resist a factored moment of 180 kNm using M25 concrete, Fe 500 steel, Width = 300 mm, Effective cover = 40 mm (Both side) Find the required areas of tension and compression reinforcement.
- | | | | | | | |
|--------------------------|---------|---------|---------|---------|---------|---------|
| Strain | 0.00174 | 0.00194 | 0.00226 | 0.00277 | 0.00312 | 0.00417 |
| Stress N/mm ² | 347.8 | 369.6 | 391.3 | 413.0 | 423.9 | 434.8 |
- B) Design a two way slab for a panel with all four edges continuous. The size of room is 3.5 m x 4.2 m. The L.L. on the slab may be taken as 2 KN/m² and F.F. 1 KN/m². Use M20, and Fe500, Use LSM.
- C) Design a square column for a working load of 1000 kN. The unsupported length of 6 column is 4.2 m. The ends of column are effectively held in position but not restrained in direction. Use M25 and Fe 500, Adopt LSM.

Q6. Solve Any Two of the following.

- A) What is an interaction curve in RCC column design? Explain its significance. 6
- B) Design a square footing for a RCC column $350 \times 350 \text{ mm}$ carrying a load of 3506 kN founded on a soil of SBC 200 kN/m² in LSM. Use M25 and Fe415 and give a neat sketch of the detailing.
- C) Design a dog legged staircase by LSM for a public building of ceiling height 3.5. 6 The width of each flight is to be kept 1.5 m. Choose suitable rise and trade. Use M20 and Fe415. Give a neat sketch of the detailing.